電気機器研究室 2025 年度 輪講資料

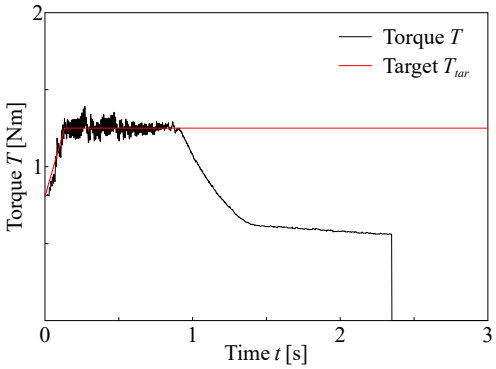
電気自動車班 研究経過報告

6 月 4 日 (水)　25M6302　荻野 健太

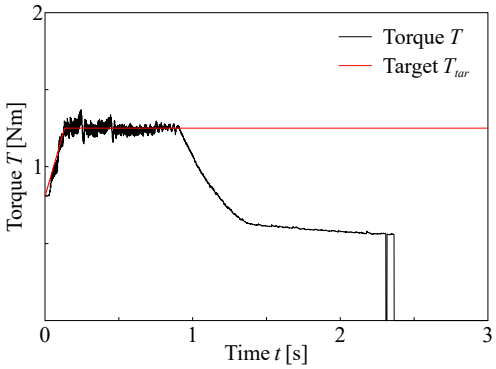
1. はじめに

2. 実験結果

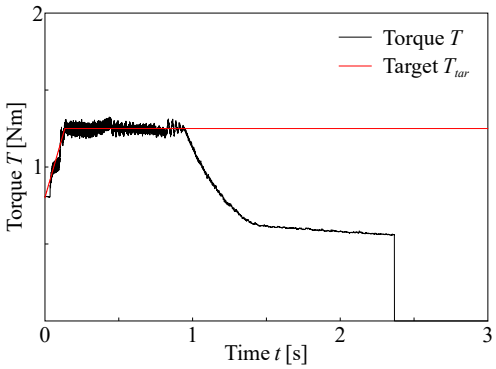
回生開始回転数N = 1000 rpm, スイッチング周波数f_{sw} = 500 Hz とし、サンプリング周波数f_sを 500 Hz から 3000 Hz まで変化させたときの回生開始時のトルク, 減速度, 回転数, デューティ比, 回生電流, 回生電圧の変化を示す。



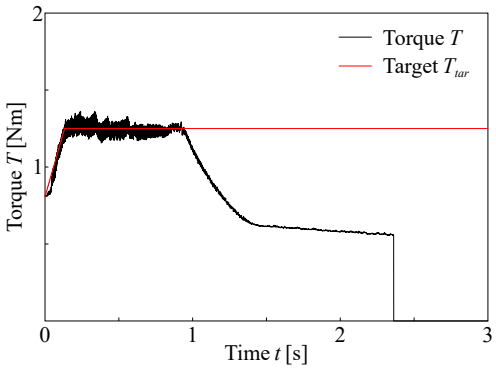
(a) $f_s = 500$ Hz



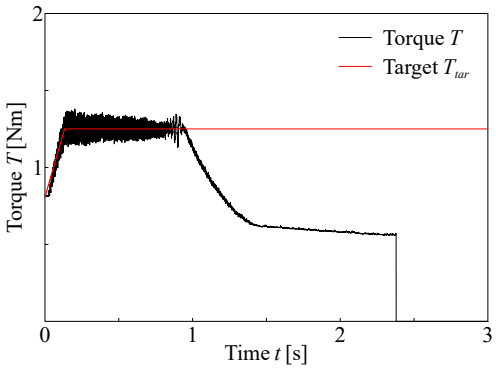
(b) $f_s = 1000$ Hz



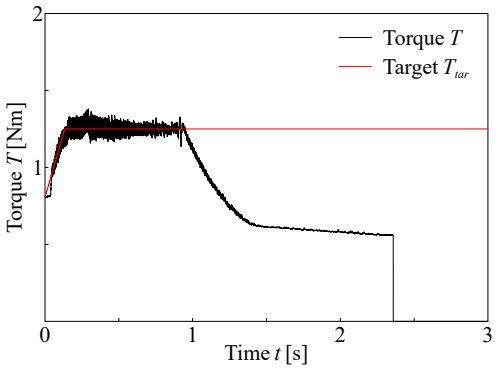
(c) $f_s = 1500$ Hz



(d) $f_s = 2000$ Hz

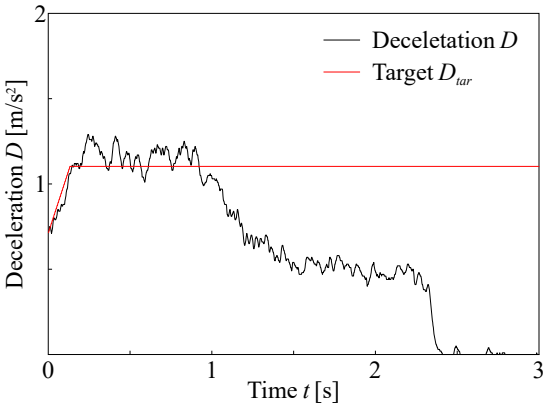


(e) $f_s = 2500$ Hz

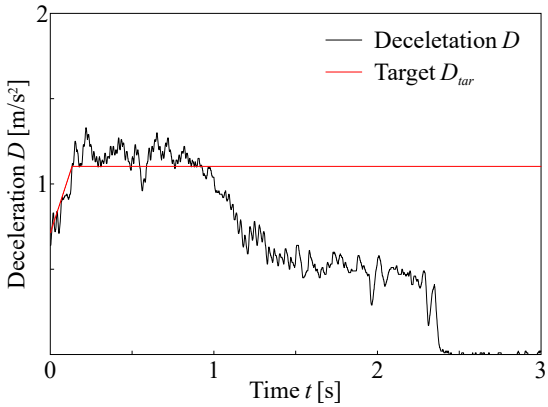


(f) $f_s = 3000$ Hz

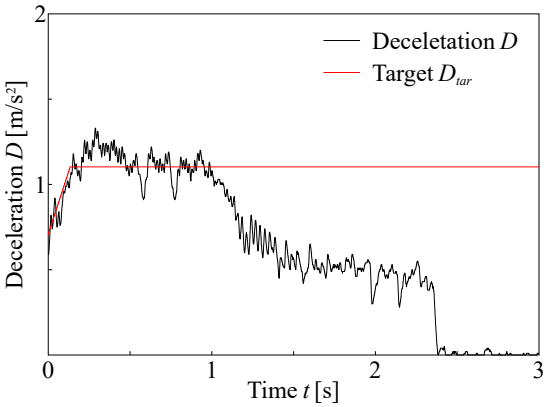
図 1　トルク



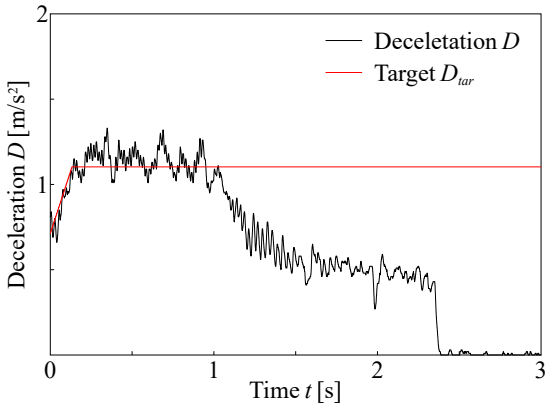
(a) $f_s = 500$ Hz



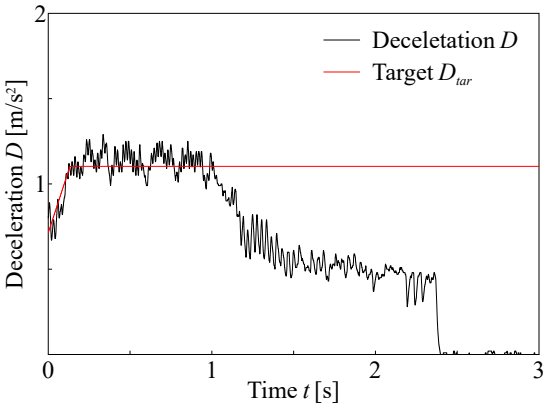
(b) $f_s = 1000$ Hz



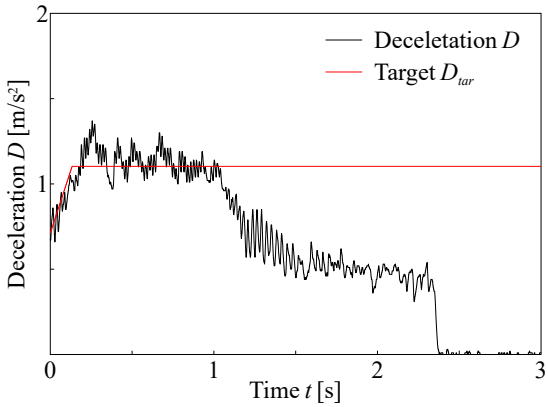
(c) $f_s = 1500$ Hz



(d) $f_s = 2000$ Hz

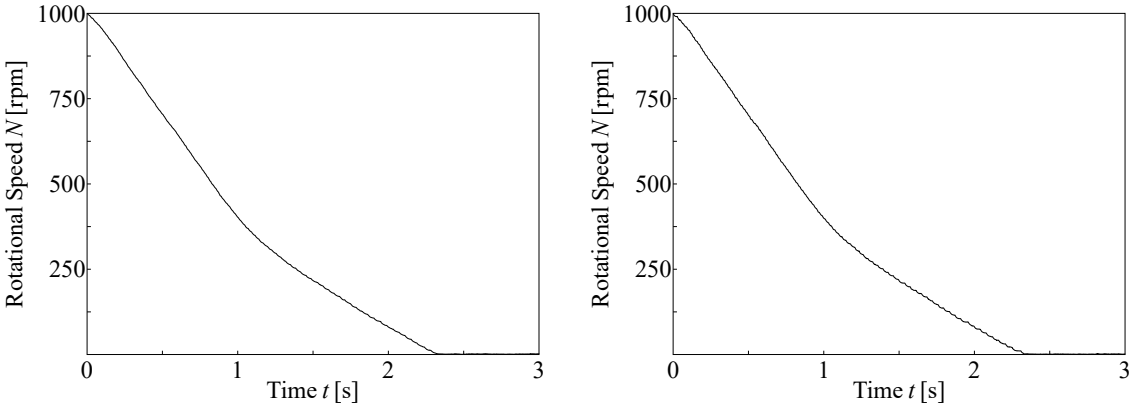


(e) $f_s = 2500$ Hz

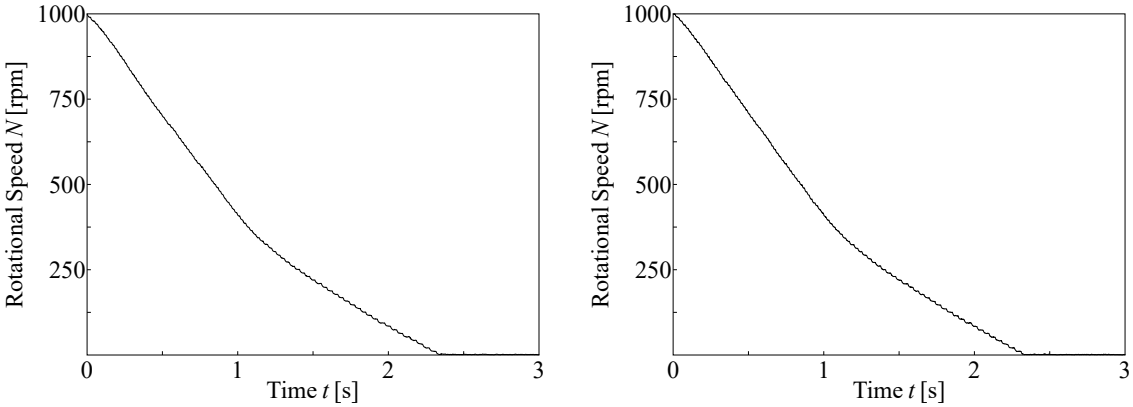


(f) $f_s = 3000$ Hz

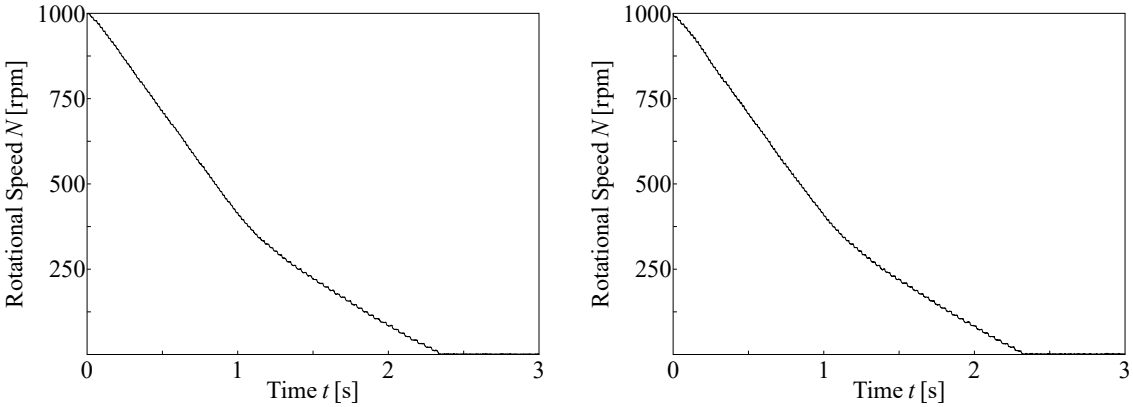
図 2　減速度



(a) $f_s = 500$ Hz　(b) $f_s = 1000$ Hz

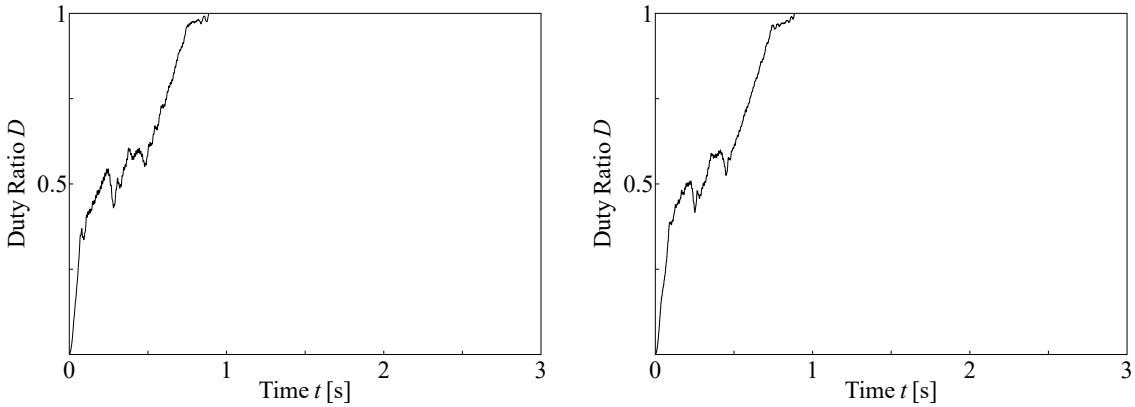


(c) $f_s = 1500$ Hz　(d) $f_s = 2000$ Hz

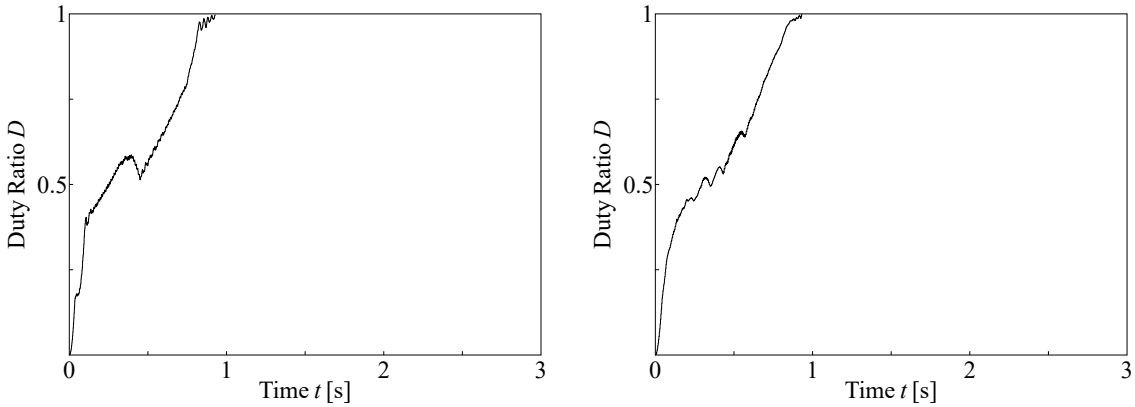


(e) $f_s = 2500$ Hz　(f) $f_s = 3000$ Hz

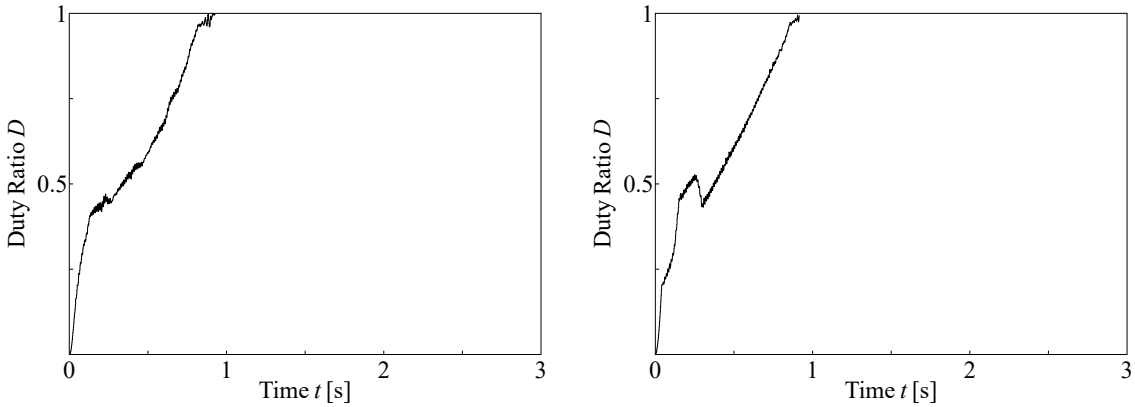
図 3　回転数



(a) $f_s = 500$ Hz　　(b) $f_s = 1000$ Hz

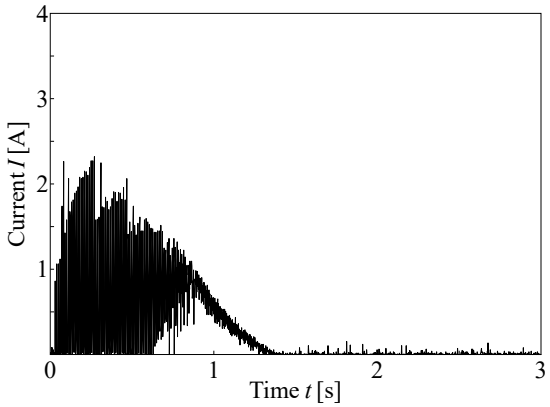


(c) $f_s = 1500$ Hz　　(d) $f_s = 2000$ Hz

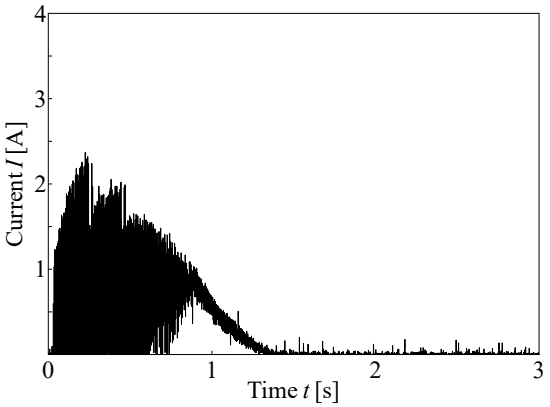


(e) $f_s = 2500$ Hz　　(f) $f_s = 3000$ Hz

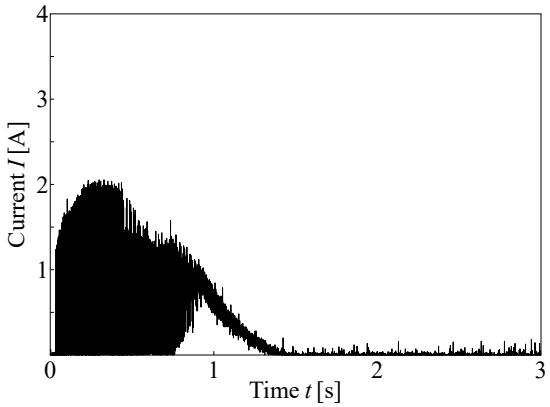
図 4　デューティ比



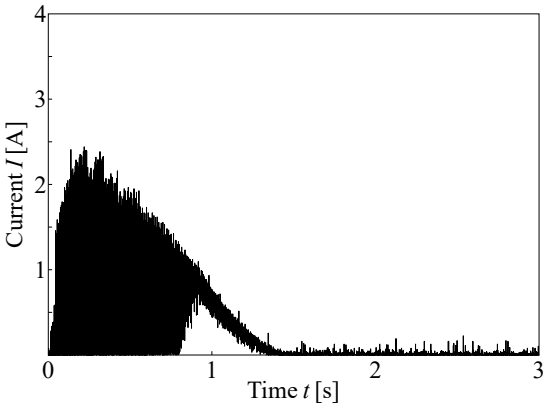
(a) $f_s = 500$ Hz



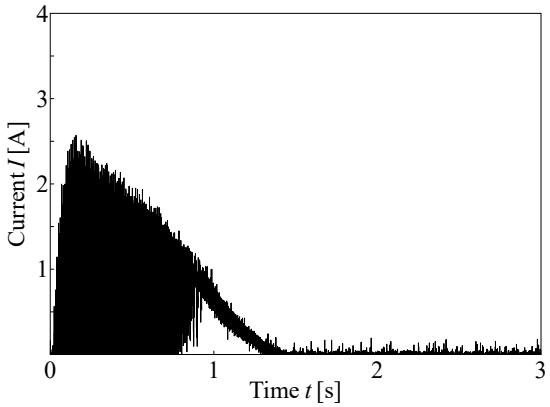
(b) $f_s = 1000$ Hz



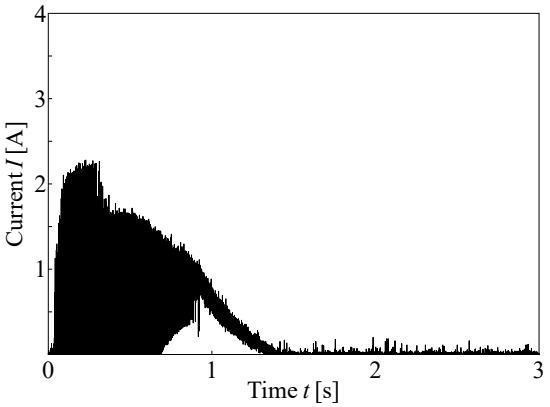
(c) $f_s = 1500$ Hz



(d) $f_s = 2000$ Hz

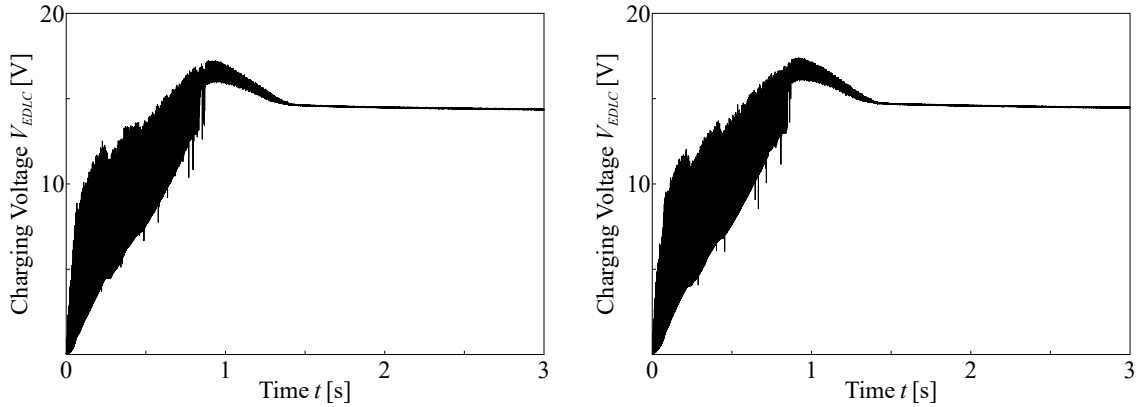


(e) $f_s = 2500$ Hz

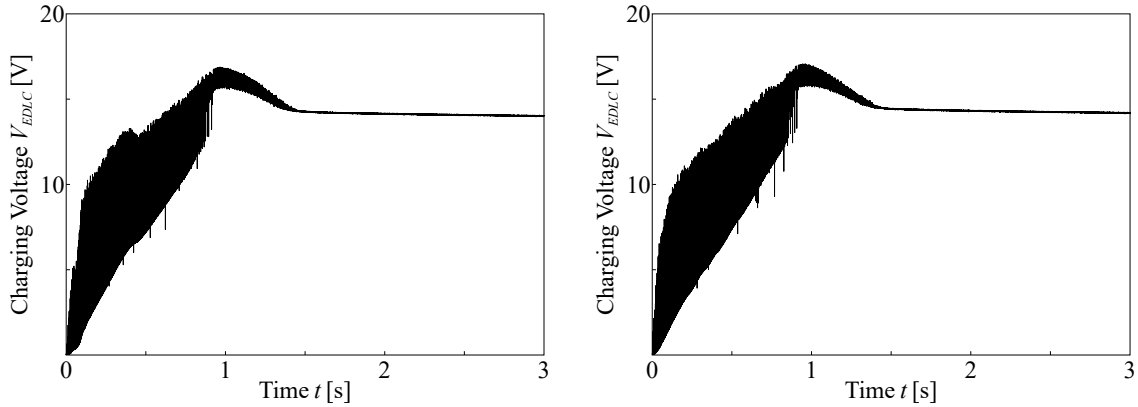


(f) $f_s = 3000$ Hz

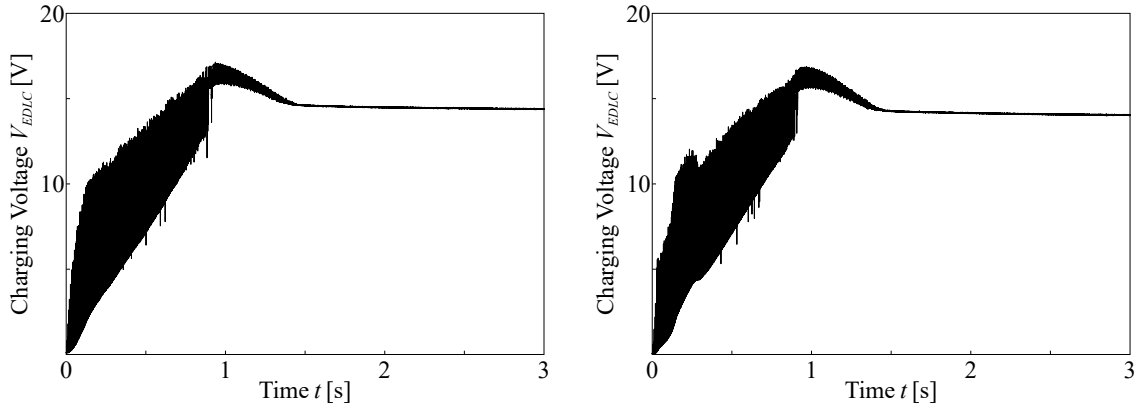
図 5　回生電流



(a) $f_s = 500$ Hz　　(b) $f_s = 1000$ Hz



(c) $f_s = 1500$ Hz　　(d) $f_s = 2000$ Hz



(e) $f_s = 2500$ Hz　　(f) $f_s = 3000$ Hz

図 6　回生電流